



ANSI/NETA ETT-2022

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STANDARD FOR

CERTIFICATION

OF ELECTRICAL TESTING TECHNICIANS

FOR ELECTRICAL POWER EQUIPMENT & SYSTEMS

NETA[®]

ANSI/NETA ETT-2022

AMERICAN NATIONAL STANDARD

**STANDARD FOR
CERTIFICATION OF
ELECTRICAL TESTING TECHNICIANS**

Secretariat
InterNational Electrical Testing Association



Approved by
American National Standards Institute



American National Standard

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Published by
InterNational Electrical Testing Association
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This standard is intended to be suitable for mandatory application by governmental bodies exercising legal jurisdiction over electrical installations and facilities; it may be used by insurance inspectors and others to codify the expertise of electrical testing personnel.

NETA Standards Review Council

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FOREWORD

(This Foreword is not part of American National Standard ANSI/NETA ETT-2022)

The InterNational Electrical Testing Association (NETA) was established in 1972 to establish uniform testing procedures for electrical equipment and apparatus. NETA developed specifications for the acceptance of new electrical apparatus prior to energization, the *ANSI/NETA Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems*, and for the maintenance of existing apparatus to determine its suitability to remain in service, the *ANSI/NETA Standard for Maintenance Testing Specifications for Electrical Power Distribution Equipment and Systems*.

In 1987, NETA recognized the need to establish a standard for technicians who perform electrical testing services. The Preface of *Standard for Certification of Electrical Testing Technicians* succinctly states the necessity of this document.

This standard was created to codify the experience, education, and training requirements necessary for an individual to obtain a level of competency as an electrical test technician. NETA carefully examines the qualifications of each NETA Accredited Company (NAC) in addition to certifying the individual technician employed by that NAC. Because of this requirement, another directive of this task group was to assure that the NETA requirements for certification were aligned with an alternate certification path for qualified individuals.

The participation of a wide variety of stakeholders during the review process contributed to the creation and acceptance of this document as an American National Standard.

ANSI/NETA Standard for the Certification of Electrical Testing Technicians was approved as an American National Standard by ANSI on March 15, 2000. On January 8, 2010, December 3, 2015, and again on February 9, 2018, ANSI/NETA ETT was approved as a revised American National Standard. The most current edition of the document, the ANSI/NETA ETT-2022, was approved as a revised American National Standard on January 7, 2022.

PREFACE

Specifying requisite levels of training, experience, and education for the evaluator of electrical power equipment is as important as the test procedure itself.

An Electrical Testing Technician (ETT) performs tests and inspections and evaluates the suitability of electrical power equipment and systems for the intended use.

Individuals performing these tests shall be capable of conducting the tests in a safe manner and with complete knowledge of the hazards involved. They must also evaluate the inspection and test data and make an assessment on the continued serviceability or non-serviceability of the equipment and systems. The evaluation of equipment requires subjective assessment by the ETT.

Certification is a means for individuals to indicate to employers, co-workers, the general public, and others that they have met the standards of an impartial, nationally-recognized organization for the performance of specific technical tasks by virtue of their technical knowledge, skills, and abilities.

CONTENTS

Page

1. Scope.....	1
2. Definitions.....	1
3. Levels of Classification.....	3
4. Training Requirements.....	5
5. Experience Requirements	6
6. Examination Requirements	7
7. Certifying Body	7

TABLES AND APPENDICES

Tables

(Supplementary information which is mandatory to the standard)

Table 1	15
Profile of Electrical Testing Technicians	
Table 2	16
Level 2 Certified Assistant Technician; Knowledge, Skill, and Ability Requirements	
Table 3	21
Level 3 Certified Technician, Knowledge; Skill, and Ability Requirements	
Table 4	27
Level 4 Certified Senior Technician, Knowledge; Skill, and Ability Requirements	

Appendices

(Supplementary information which is not mandatory to the standard)

Appendix A.....	33
Experience and Training Verification by Employer	
Appendix B.....	36
References	

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NETA STANDARD FOR CERTIFICATION OF ELECTRICAL TESTING TECHNICIANS

1. SCOPE

1. This standard establishes minimum requirements for qualification and certification of the Electrical Testing Technician (ETT).
2. This standard details the minimum training and experience requirements for Electrical Testing Technicians and provides criteria for documenting qualifications and certification.
3. This standard details the requirements for an independent and impartial certification system to certify Electrical Testing Technicians.

2. DEFINITIONS and ACRONYMS

The definitions in this standard are intended to clarify the meaning of terms used in this standard, as they apply to this standard, and only to this standard. No broader application of these definitions is implied.

Appeal

Request by applicant, candidate, or certified person for reconsideration of any adverse decision made by the certification body related to his/her desired certification status.

Appendix

Supplementary information which is informational but not mandatory to this standard.

Candidate

Individual who does not hold a current and valid certificate but may be involved in a certification system as an applicant or examinee.

Certificant

Individual who holds a current and valid Electrical Testing Technician (ETT) certificate.

Certification Body (Certifying Body)

Organization that meets the criteria as outlined in this standard for offering an Electrical Testing Technician (ETT) certification program.

Certification Element

An area of job performance on which a candidate for certification will be tested by written examination.

Certification Process

The process by which a third party provides written assurance that a person's competence conforms to specified requirements.

Certification Program

Compilation of scope of certification, job and task description, required competence, abilities, and prerequisites.

Certification System

Rules and procedures governing the certification program.

Certify

To confirm formally that a candidate has met the criteria and passed a comprehensive written examination that indicates competence in electrical testing.

Closed-Book Examination

An examination administered without access to reference material except that supplied with or in the examination.

Documented

The condition of being in written form.

Education

An institutionalized program, prescribed by appropriate authorities, that is offered by schools, institutes, organizations, colleges, or universities established for the sole purpose of providing instruction in an orderly, planned, and systematic fashion.

Electrical Testing Technician (ETT)

Acronym for an individual who performs tests and inspections and evaluates the suitability of electrical power equipment and systems for the intended use.

Employer

The corporate, private, or public entity that employs personnel for wages or salary.

Evaluation

Process that assesses a person's fulfillment of the certification system requirements, leading to a decision on certification.

Examination

Mechanism that is part of the evaluation, which measures a candidate's competence with regard to the procedures, codes, standards, specifications, technical knowledge, and equipment or instrumentation for electrical testing. Competence may be measured by one or more means such as computer-based, written, oral, practical, and observational methods.

Experience

Actual performance or observation conducted during work time resulting in the acquisition of knowledge, skills, and abilities.

KSA

Acronym for knowledge, skills, and abilities.

Method

A systematic and regular means of accomplishing a given task.

Proctor

Person who administers the examination component of a certification system.

Psychometrics

Science and technology of mental measurement.

Qualification

The education, training, knowledge, skills, abilities, and experience required for an individual to perform at a specified ETT level.

Quality Control

Operational techniques and activities that are used to fulfill requirements of quality.

RLC

Abbreviation for resistive, inductive, capacitive.

Stakeholder

Party with a direct interest in the use of a certification system, including candidates, certificants, employers, authorities, and specifying entities.

Training

An organized and documented program of activities designed to impart the knowledge and develop the skills and abilities required by this standard. On-the-job experience does not qualify as training.

3. LEVELS OF CLASSIFICATION

Four levels of classification are defined in terms of the knowledge, skills, and abilities required in a given method or methods to perform specified electrical testing and maintenance activities. To proceed to a higher level, an individual shall meet the qualifications set forth and pass the examination for all previous levels.

1. ETT Trainee Level 1

The ETT Trainee Level 1 classification offers an entry-level gateway into the electrical testing industry. Typical duties include providing assistance to higher level technicians, pre- and post-test sequence assembly and disassembly, and performing simple measurements and/or tests under direct supervision

1. The candidate has a high school or general equivalency diploma (GED).
2. The candidate has a minimum of documented 8 hours of electrical safety training.
3. There are no requirements for related work experience in the electrical testing industry.
4. Table 1 contains specific requirements.

2. ETT Certified Assistant Level 2

The ETT Certified Assistant Level 2 performs limited testing and service work while generally requiring direct supervision. The individual has sufficient knowledge and experience to be qualified for assuring their own safety. Safety knowledge includes an understanding of lockout/tagout procedures and requirements, arc-flash and shock risk assessments, and other facets of hazardous electrical energy control procedures.

1. The candidate shall have two years of related experience in the electrical testing industry. This experience shall be verified by the employer using a form similar to the one in Appendix A.
2. The candidate shall have a minimum of 40 hours training in safety. This training shall be verified by the employer using a form similar to the one in Appendix A.
3. The candidate shall have a minimum of 160 hours of training related to the certification element categories. This training shall be verified by the employer using a form similar to the one in Appendix A.
4. The candidate shall pass an Electrical Test Technician Level 2 examination administered by a certifying body as defined in this standard.
5. Tables 1 and, 2 contain specific requirements. Requirements for each level of certification are inclusive of previous levels.

3. ETT Certified Level 3

The ETT Certified Level 3 is capable of supervising ETT Trainee Level 1 and ETT Certified Assistant Level 2. Typical duties include, but are not limited to, performance and management of routine and moderately complex tasks and projects, record keeping, evaluation of test data, and responsibility for the safety of others. Individual is qualified to provide guidance and is capable of performing electrical power switching.

1. The candidate shall have a minimum of five years of full-time experience in the electrical testing industry. This experience shall be verified by the employer using a form similar to the one in Appendix A.
2. The candidate shall have a minimum of 24 hours of additional documented safety training. This training shall be verified by the employer using a form similar to the one in Appendix A.
3. The candidate shall have 240 hours of additional documented training related to the certification element categories for Level 3. This training shall be verified by the employer using a form similar to the one in Appendix A.
4. The candidate shall pass an Electrical Test Technician Level 3 examination administered by a certifying body as defined in this standard.
5. Tables 1, 2, and 3 contain specific requirements. Requirements for each level of certification are inclusive of previous levels.

4. ETT Senior Certified Level 4

The ETT Senior Certified Level 4 is capable of supervising large projects and multiple crews and can work independently. Individual performs complex investigations, tests, and evaluations, and prepares written reports as needed.

1. The candidate shall have a minimum of ten years of full-time experience in the electrical testing industry. This experience shall be verified by the employer using a form similar to the one in Appendix A.
2. The candidate shall have a minimum of 40 hours additional safety training. This training shall be verified by the employer using a form similar to the one in Appendix A.
3. The candidate shall have a minimum of 200 hours of additional training related to the certification element categories for Level 4. This training shall be verified by the employer using a form similar to the one in Appendix A.
4. The candidate shall pass an Electrical Test Technician Level 4 examination administered by a certifying body as defined in this standard.
5. Tables 1, 2, 3, and 4 contain specific requirements. Requirements for each level of certification are inclusive of previous levels.

4. TRAINING REQUIREMENTS

1. ETT Trainee Level 1

1. The candidate shall complete sufficient training to become familiar with the principles of the industry, the practices of applicable electrical testing techniques, and necessary safety requirements applicable to Level 1.
2. Representative training curricula shall include, but not be limited to, the subjects contained in Table 1.

2. ETT Certified Assistant Level 2

1. The candidate shall complete sufficient training to become familiar with the principles of the industry, the practices of the applicable electrical testing techniques, and necessary safety requirements applicable to Level 2.
2. Representative training curricula shall include, but not be limited to, the subjects contained in Tables 1, and 2.

3. ETT Certified Level 3

1. The candidate shall complete sufficient training to enhance his/her knowledge and skills in the area of electrical testing and necessary safety requirements applicable to Level 3.

2. ETT certified technicians are “hands-on” field testing technicians. These technicians shall be trained concerning the hazards, components, nominal voltages, operating characteristics, and limitations of electrical power and electrical power systems.
 3. ETT certified individuals shall have knowledge in applied mathematics, physical science, and engineering science equivalent to that provided by course work and hands-on laboratory work of an associate degree program in electrical engineering technology. A degree is not a requirement for certification.
 4. Representative training curricula shall include, but not be limited to, the subjects contained in Tables 1, 2, and 3.
4. ETT Senior Certified Level 4
1. The candidate shall complete sufficient organized training to enhance his/her knowledge and skills in the area of electrical testing and necessary training requirements applicable to Level 4.
 2. ETT senior certified technicians are “hands-on” field testing technicians. These technicians shall be trained concerning the hazards, components, nominal voltages, operating characteristics, and limitations of electrical power and electrical power systems.
 3. Training shall include all topics contained in Tables 1, 2, 3, and 4 for the appropriate level plus advanced topics that are directly applicable to electrical testing as deemed necessary by the employer.

5. EXPERIENCE REQUIREMENTS

Candidates shall have acquired the practical experience to assure they are competent to perform the duties at the level of classification which is being sought. Complete experience requirements are set forth in Table 1.

1. ETT Trainee Level 1

The candidate shall have a high school or GED degree.
2. ETT Certified Assistant Level 2

The candidate shall have related experience assisting in testing of electrical power systems for two years.
3. ETT Certified Level 3

The candidate shall have been engaged on a full-time basis performing testing of electrical power systems for five years.
4. ETT Senior Certified Level 4

The candidate shall have been engaged on a full-time basis performing testing of electrical power systems for ten years.

6. EXAMINATION REQUIREMENTS

1. ETT Trainee Level 1

Competency of the candidate shall be determined by the employer to confirm the knowledge and skills outlined in Table 1 and Appendix A.

2. Examination for ETT Certified Assistant Level 2, Certified Technician Level 3, Senior Certified Technician Level 4.

1. Examinations shall be prepared by a certifying body as defined in this standard to test for the knowledge and skills outlined in Tables 1, 2, 3, and 4, as applicable, and Appendix A.
2. Examinations shall be administrated by a certifying body as defined in this standard.

7. CERTIFYING BODY

This standard defines the requisite certification criteria for an independent, third-party evaluator. Implicit in this document is that certifying bodies shall maintain and improve their certification systems in accordance with the following general provisions.

1. General Provisions

1. The certifying body for Electrical Testing Technicians shall be a recognized evaluator in the area of electrical testing.
2. The certifying body shall detail its policies and procedures in documents available in electronic or print format. These documents shall be reviewed on a regular basis with a record of such review maintained.
3. The policies and procedures under which the certification body operates and its administration shall be nondiscriminatory and shall be administered in a nondiscriminatory manner. Procedures shall not be used to impede or inhibit access by qualified applicants.
4. The certifying body shall not be owned in whole or in part by the company which employs the certified individual.
5. The certification body shall make its services accessible to all qualified applicants. There shall not be undue financial or other conditions. Access shall not be based on biased conditions such as required training by a specific organization or a specific group of organizations; however, training recognized by the certification body may be required. Organizations may have specific requirements involving ethics, fiduciary responsibility, malfeasance, etc. and, under these

circumstances, may require membership in the organization as a condition of certification.

6. Certification shall not be conditional upon the number of persons already certified.
7. The certification body shall clearly define qualification criteria.
8. The certification body shall provide (through documents, electronic media, or other means), update periodically, and make available upon request:
 1. A documented statement of its certification system, including its policies and procedures for granting, maintaining, extending, restricting, suspending, and withdrawing certificates.
 2. Information about the examination and certification procedure, including a detailed description of technical and other requirements.
 3. A policy concerning the rights and responsibilities of applicants and certificants as specified, including requirements, restrictions, or limitations on the use of the certification body's marks and on the ways of referencing the certificants granted.
 4. Policy and procedures for handling complaints, appeals, and disputes.
 5. Policy and procedures to act on documented cases of certificants' inappropriate or incompetent work or on cases in which certificants utilize certificates in an inappropriate or misleading manner. *Note:* This may include corrective measures, publishing the violation, and, if necessary, legal action.
9. The certification body shall verify, upon written request, the scope and validity of certifications granted.
10. The certification body may establish audit procedures for verification of information provided by employers, candidates, and/or technicians.

2. Organization

The certifying body shall:

1. Be a third-party, corporately independent organization which can function as an unbiased certification authority.
2. Be responsible for its decisions relating to the granting, maintaining, extending, restricting, suspending, and withdrawing of certificates issued.
3. Be financially stable.
4. Have a documented quality control system.

5. Ensure that examination and certification are performed independent of training services.
6. Ensure that examination and certification are performed independent of the employer.
7. Establish a committee consisting of subject matter experts (SME) with technical expertise and diversity of field experience in electrical testing. This committee is responsible for technical matters relating to the ETT examination.
8. Conduct periodic internal audits covering all procedures in a planned and systematic manner.

3. Subcontracting

When a certification body subcontracts work such as exam proctoring to an external body or person, the certification body shall draw up an appropriately documented agreement which includes confidentiality and prevention of conflict of interest. The certification body shall:

1. Take full responsibility for subcontracted work and maintain its responsibility for granting, extending, suspending, and withdrawing certificates.
2. Ensure that the subcontracted body or person is competent and complies with the applicable provisions of this standard and is not involved, directly or through an employer, with any activities that could result in compromising their impartiality.

4. Conditions for granting, maintaining, extending, suspending, and withdrawing certification

1. The certification body shall specify the conditions and procedures for granting, maintaining, and restricting certification and the conditions under which certificates may be suspended or withdrawn for all or part of a certificate's scope.
2. The certification body shall require certified persons to act in accordance with the certification body's published policies and procedures and other normative documents.

5. Records

1. The certification body shall maintain a record system to suit its particular circumstances and to comply with existing regulations. The records shall demonstrate that certification procedures have been appropriately applied, particularly in respect to application forms, examination documentation and evaluations, and other documents relating to granting, maintaining, extending, restricting, suspending, or withdrawing certification.
2. Records shall be identified, managed, and disposed of in such a way as to ensure confidentiality and to protect the data contained in the procedures described in such records.

3. The certification body shall have policy and procedures for retaining records for a period consistent with its contractual, legal, or other obligations. The records shall be kept for a sufficient period of time so that continued confidence may be demonstrated for at least one full certification cycle or as required by law.
4. The certification body shall have policy and procedures concerning access to these records in accordance with this standard.

6. Confidentiality

1. The certification body shall have adequate arrangements, consistent with applicable laws, to safeguard confidentiality of the information obtained in the course of its certification activities at all levels of its organization, including committees and external bodies or persons acting on its behalf.

7. Examination

Examination(s) shall be based on psychometrically sound principles.

1. Examination Facility

1. Examinations shall be conducted at a commercial testing facility or an accredited institution of higher education such as a college, junior college, or university.
2. The examination shall be administered in a secure area that provides the environmental and physical conditions that are generally accepted as necessary for an examination facility.
3. The candidate, his employer, or associates shall not contact the examination facility administration personnel or proctor, with the exception of scheduling and taking the certification examination.

2. Proctors

1. Proctors shall be impartial and third-party, with experience in supervising examinations.
2. Proctors do not need and, preferably, shall not have experience in performing or supervising the performance of the tasks covered by the examination.
3. The certifying body shall provide complete instructions detailing duties and responsibilities of the proctor.
4. The primary responsibility of the proctor is to maintain the integrity of the examination process. Duties include but are not limited to:
 - a. Verification of the identity of candidate(s).

- b. Distribution and collection of examinations and all printed matter.
- c. Monitoring candidates during the examination.
- d. Maintaining security of all materials prior to, during, and following the examination.

3. Security

The certification body shall implement policies and procedures concerning security in order to:

- 1. Maintain security of all examination questions, all versions of the examination, and other examination materials.
- 2. Ensure the confidentiality of candidates and certificants' demographic information, work experience record, exam results, other personal information.

4. Process

The files of the certification body shall include the following information.

- 1. Rationale for the certification program.
- 2. Exam specifications which shall include:
 - a. Content outline based on Tables 1, 2, 3, and 4.
 - b. Types of exam questions used.
 - c. Time constraints of the examination(s).
 - d. Appropriate cognitive levels of exam questions.
 - e. Scoring procedures.
 - f. Acceptable statistical characteristics of exam questions.
- 3. Exam question content, source, and validation history.
- 4. Data on each examination to include content, date(s) administered, and validation history.
- 5. Compilation of data to evaluate the validity, reliability, and general performance of all individual exam questions and all versions of an examination.
- 6. Demographic information, examination records, work history, and certification status for all candidates and certificants.

7. Identity of proctor and exam location.
5. Content
 1. All versions of the exam shall conform to the exam specification.
 2. All versions of the exam shall be changed on a regularly-scheduled basis to maintain security and to keep current with technical changes within the industry.
 3. The certification body shall have procedures to ensure that no candidate receives the same version of the exam twice.

6. Re-examination

The certification body shall establish policy and procedure for re-examination, which includes a waiting period to allow candidates sufficient time to learn the appropriate procedures and apply them to the job.

8. Changes in the certification requirements

The certification body shall give due notice to representatives of interested parties of any changes in its requirements for certification. It shall take into account views expressed by interested parties before deciding on the precise form and effective date of the changes. Following decisions on, and publication of, the changed requirements, the certification body shall inform the interested parties appropriately.

9. Appeals, complaints, disputes

Candidates shall be informed of:

1. The extent to which they may appeal their eligibility determination and review their examination performance.
2. The process for filing a claim of examination error or other challenge, including what may be expected from the process and outcome.
3. Re-examination policies and procedures.

10. Verification of Eligibility

1. The certification body shall require the verification of eligibility of the applicant seeking certification. This verification shall include the following:
 1. Certification classification.
 2. Details of relevant qualifications, confirmed and supported by evidence. (i.e., training and work experience).

2. The following minimum information shall be provided by the applicant prior to the qualification examination:
 1. General information on the applicant, e.g., name, address, and other information required to identify the person.
 2. A statement that the person agrees to comply with the requirements for certification and to supply any information needed for the evaluation.
 3. Signature verifying that the applicant agrees to comply with the requirements for certification.
3. The information gathered shall be treated with the appropriate confidentiality by the certification body.

11. Certification

1. The certification body shall provide to all certified persons certificates signed by an officer assigned such responsibility. These certificates shall contain the following information:
 1. Name of the certificant.
 2. Certification or identification number of the certificant.
 3. Name of certifying body.
 4. Level and title of certification.
 5. Effective date of certification, including expiration date.
2. Certificate shall be valid for a specific time period, not to exceed three years.
3. The certification body shall establish a procedure for recertification through continuing technical development, re-examination, or a combination thereof.
4. Certificants shall be required to perform duties in an ethical, responsible, and competent manner.
5. Certificants shall be required to inform the certifying body of changes in their status that may impact their certification status.

12. Use of certificates and marks

1. The certification body, which is the owner of the certification mark, shall appropriately manage and monitor the rights for usage and representation.
2. If the certification body grants the right to use a mark or logo, the certified person may use the specified mark or logo only according to the certification body's conditions as specified in writing.

3. Inappropriate references to the certification system or misleading use of certificates and marks in publications, catalogs, etc., shall be answered with suitable measures.

Note: This could include corrective measures, withdrawal of certification, publication of the infraction, and, as appropriate, additional legal action.

13. Self-Certification Programs

Self-certification programs do not meet the criteria set forth in this standard since an organization providing self-certification has a significant financial stake in the outcome of the certification process. Self-certification does not provide independent, third-party objectivity.

Table 1
Profile of Electrical Testing Technicians

TITLE: LEVEL:	Trainee Technician Level 1	Certified Assistant Technician Level 2	Certified Technician Level 3	Certified Senior Technician Level 4
EDUCATION AND TECHNICAL TRAINING:	High School / GED	Electrical 160 hours	Electrical 240 hours	Electrical 200 hours
SAFETY TRAINING:	8 hours [electrical]	40 hours	24 hours	40 hours
RELATED EXPERIENCE:	None	Two Years*	Five Years*	Ten Years*
TYPICAL ABILITY:	Requires supervision	Requires limited supervision	Lead technician	Supervisor
TYPICAL DUTIES:	Responsible for safety of self.	Understands hazardous electrical energy control procedures.	- Supervises Levels I and II. - Routine and moderately complex projects. - Record keeping. - Safety of others. - Switching. - Evaluations.	- Supervises large projects, multiple crews. - Works independently. - More complex investigations, tests, and evaluations.
TYPICAL ACTIVITIES:	- Assists. - Basic measurements. - Test equipment set up and removal. - Basic maintenance and cleaning.	- Basic visual and mechanical inspections - Basic Tests. - Data collection. - Test for de-energized locked out/tagged out equipment.	- Lockout/tagout, safety grounding. - Advanced visual and mechanical inspections and tests. - Interacts with other skills and operations. - Writes reports. - Troubleshooting.	- Recommends Corrective actions. - Complex visual and mechanical inspections and tests. - Interacts with engineers and managers. - System Commissioning.
EXAMINATION:	By employer	By certification body	By certification body	By certification body
NOTE: Candidates must have met the qualifications and training requirements for all previous levels. * Completion of two or more years of technical education in an electrical field shall be equivalent to a maximum of one year of experience.				

Table 2
Level 2 Certified Assistant Technician
Knowledge, Skill, and Ability Requirements

NOTE: Where a knowledge, skill, or ability (KSA) exists for multiple certification levels, it is intended that each successive level will possess more in-depth of knowledge of that KSA.

1. Safety

A. Risk Assessment

1. Identify safe work practices with regard to hazards and associated risk with respect to assigned tasks.
2. Determine the location and implications of shock and arc-flash protection boundaries

B. Establishing an Electrically Safe Work Condition

1. Apply requirements of NFPA 70E, Article 120: *Establishing an Electrically Safe Work Condition*

C. Lockout/Tagout Procedures

1. Apply OSHA lockout/tagout requirements

D. Personal Protective Equipment (PPE)

1. Understand Arc-Rated PPE (incident energy ratings) and PPE for Shock Protection
2. Apply correct use, storage, inspection, and testing requirements of PPE

E. Safety Equipment Selection

1. Identify the safety equipment needs of the tasks to be performed (e.g., hotstick, voltage detectors, insulated tools, fire extinguisher, etc.)

F. Confined Space

1. Recognize confined space and its hazards
2. Recognize when a confined space becomes a permit-required confined space along with the associated requirements
3. Apply OSHA requirements

G. Switching and Temporary Protective Grounding

1. Select and apply correct voltage detection equipment
2. Apply correct temporary protective grounding equipment
3. Interpret and apply system switching procedures

H. Incident Energy Analysis

1. Interpret field-marked equipment labels
2. Identify hazards where no field-marked labels exist
3. Demonstrate knowledge of various methods used to determine incident energy
4. Demonstrate knowledge of how distance, current, time, and switchgear configuration affect incident energy
5. Demonstrate basic understanding of Arc Energy Reduction techniques and testing requirements in the NEC

I. Codes and Standards

1. Recognize the scope of standards and regulatory organizations (e.g., OSHA, ANSI, ASTM, IEEE, NETA, NFPA, NEMA, CSA, etc.)
2. Recognize key provisions of NFPA 70 (e.g., Articles 90, 110.16, 110.21, 225.56, 230.95, 250, and 310) and 70E (i.e., Chapter 1: Safety-Related Work Practices and Chapter 2: Safety-Related Maintenance Requirements, etc.)
3. Apply basic technical requirements of ANSI/NETA Standards to the assigned task

Table 2
Level 2 Certified Assistant Technician
Knowledge, Skill, and Ability Requirements

4. Demonstrate an understanding of how manufacturer's instructions and listing and labeling information impacts electrical testing and maintenance

2. Electrical Testing Fundamentals and Theory

A. Fundamentals of Electricity

1. Define and apply basic terms and concepts from physics, electricity, thermodynamics, and chemistry
2. Recognize and define the standard units used to describe electrical circuits, energy, and power
3. Identify series and parallel circuits
4. Recognize the sources and common effects of electromagnetic fields
5. Demonstrate knowledge of dielectric properties of various types of insulations
6. Demonstrate knowledge of phasing and phase rotation

B. Electrical Calculations

1. Perform mathematical calculations utilizing basic algebra, geometry, and trigonometry
2. Apply fundamental electrical laws to simple circuits (e.g., Ohm's law, Kirchhoff's law, etc.)
3. Apply conversions between units (e.g., microhms to milliohms, horsepower to kilowatts, etc.)

C. AC and DC Circuits

1. Calculate simple single-loop ac circuits with RLC components
2. Calculate equivalent resistance of and power consumed by resistive circuits
3. Calculate equivalent capacitance and inductance
4. Apply methods for measuring and calculating impedance and power
5. Demonstrate knowledge of basic ac and dc circuits (e.g., voltage drop, current flow, power, reactance, etc.)
6. Demonstrate knowledge of phase angle, power factor, and vectors

D. Insulation Testing

1. Employ methods and procedures for basic insulation tests and analyze results

E. Resistance Testing

1. Employ methods and procedures for basic resistance tests and analyze results (e.g., winding resistance, contact resistance, bolted connections, etc.)

F. Thermographic Survey

1. Employ methods and procedures for thermographic surveys and identify deficiencies (e.g., emissivity, reflection, delta-T, etc.)

G. Current Testing

1. Employ methods and procedures for basic current tests and analyze results (e.g., overcurrent pickup, circuit burden, voltage drop, etc.)

H. System Tests, Analysis, and Operation

1. Demonstrate understanding of electrical one-line diagrams
2. Interpret ac and dc schematic diagrams, ac three-line diagrams, and connection and interconnection drawings
3. Identify electrical symbols and ANSI device numbers
4. Interpret time-current curves

Table 2
Level 2 Certified Assistant Technician
Knowledge, Skill, and Ability Requirements

5. Recognize essential components of short-circuit and coordination studies (e.g., cable size and length, available protective device settings, raceway type, utility fault contribution, etc.)
6. Demonstrate basic understanding of short-circuit and coordination studies

3. Component Testing

A. Switchgear and Switchboard Assemblies

1. Identify properties, types, and applications
2. Apply visual and mechanical inspections
3. Employ methods and procedures for electrical tests
4. Evaluate results of basic electrical tests

B. Transformers

1. Identify properties, types, and applications
2. Identify properties, types, and applications of auxiliary devices
3. Apply visual and mechanical inspections
4. Employ methods and procedures for basic electrical tests
5. Evaluate results of basic electrical tests

C. Cables

1. Identify properties, types, and applications
2. Identify properties, types, and applications of accessories
3. Apply visual and mechanical inspections
4. Employ methods and procedures for basic electrical tests
5. Evaluate results of basic electrical tests
6. Demonstrate knowledge of cable fault locating

D. Metal-Enclosed Busways

1. Identify properties, types, and applications
2. Apply visual and mechanical inspections
3. Employ methods and procedures for basic electrical tests
4. Evaluate results of basic electrical tests

E. Switches

1. Identify properties, types, and applications
2. Apply visual and mechanical inspections
3. Employ methods and procedures for electrical tests
4. Evaluate results of basic electrical tests

F. Circuit Breakers

1. Identify properties, types, and applications
2. Apply basic visual and mechanical inspections
3. Employ methods and procedures for basic electrical tests
4. Evaluate results of basic electrical tests

G. Circuit Switchers

1. Identify properties, types, and applications
2. Apply visual and mechanical inspections
3. Employ methods and procedures for basic electrical tests
4. Evaluate results of basic electrical tests

H. Network Protectors

1. Identify properties, types, and applications

I. Protective Relays

Table 2
Level 2 Certified Assistant Technician
Knowledge, Skill, and Ability Requirements

1. Identify properties, types, and applications of basic protective relays
 2. Apply visual and mechanical inspections
 3. Employ methods and procedures for electrical tests for basic protective relays-overcurrent, over/under voltage, frequency, etc.
 4. Evaluate test results of basic protective relays
- J. Instrument Transformers
1. Identify properties, types, and applications
 2. Apply visual and mechanical inspections
 3. Employ methods and procedures for basic electrical tests
 4. Evaluate results of basic electrical tests
- K. Metering Devices
1. Identify properties, types, and applications
 2. Apply visual and mechanical inspections
 3. Employ methods and procedures for basic electrical tests
 4. Evaluate results of basic electrical tests
- L. Regulating Apparatus
1. Identify properties, types, and applications
 2. Apply basic visual and mechanical inspections
- M. Grounding Systems
1. Identify properties, types, and applications
 2. Apply visual and mechanical inspections
 3. Employ methods and procedures for basic electrical tests
 4. Evaluate results of basic electrical tests
- N. Ground-Fault Protections Systems
1. Identify properties, types, and applications
 2. Apply visual and mechanical inspections
 3. Employ methods and procedures for basic electrical tests
 4. Evaluate test results for basic systems
- O. Rotating Machinery
1. Identify properties, types, and applications
 2. Apply visual and mechanical inspections
 3. Employ methods and procedures for electrical basic tests
 4. Evaluate results of basic electrical tests
- P. Motor Control Centers and Motor Starters
1. Identify properties, types, and applications
 2. Apply visual and mechanical inspections
 3. Employ methods and procedures for basic electrical tests
 4. Evaluate results of basic electrical tests
- Q. Adjustable-Speed Drive Systems
1. Identify properties, types, and applications
- R. Direct-Current Systems
1. Identify properties, types, and applications
 2. Apply visual and mechanical inspections
 3. Employ methods and procedures for basic electrical tests
 4. Evaluate results of basic electrical tests
- S. Surge Arresters

Table 2
Level 2 Certified Assistant Technician
Knowledge, Skill, and Ability Requirements

1. Identify properties, types, and applications
 2. Apply visual and mechanical inspections
 3. Employ methods and procedures for basic electrical tests
 4. Evaluate results of basic electrical tests
- T. Capacitors and Reactors
1. Identify properties, types, and applications
 2. Apply visual and mechanical inspections
 3. Employ methods and procedures for basic electrical tests
 4. Evaluate results of basic electrical tests
- U. Outdoor Bus Structures
1. Identify properties, types, and applications
 2. Apply visual and mechanical inspections
 3. Employ methods and procedures for basic electrical tests
 4. Evaluate results of basic electrical tests
- V. Emergency Systems
1. Identify properties, types, and applications
 2. Apply visual and mechanical inspections
 3. Employ methods and procedures for basic electrical tests
 4. Evaluate results of basic electrical tests
- W. Automatic Circuit Reclosers and Line Sectionalizers
1. Identify properties, types, and applications
 2. Apply visual and mechanical inspections
- X. Fiber-Optic Cables
1. Recognize and correctly handle fiber-optic cables
- Y. Insulating Liquids and Gases
1. Identify properties, types, and applications
 2. Apply sampling procedures in accordance with ASTM
 3. Demonstrate understanding of ASTM dielectric breakdown tests
 4. Evaluate results of basic ASTM tests
- Z. Fuses
1. Identify properties, types, and applications
 2. Apply visual and mechanical inspections
 3. Employ methods and procedures for electrical tests
 4. Evaluate results of basic electrical tests

4. Systems and Commissioning

- A. Troubleshooting
1. Investigate power or protective system simple faults or malfunctions to determine cause and corrective action required
- B. Functional Testing
1. Employ methods and procedures for basic system-function tests upon completion of the individual component tests defined, as system conditions allow
 2. Apply concepts of functionality for simple electrical systems

Table 3
Level 3 Certified Technician
Knowledge, Skill, and Ability Requirements in Addition to Level 2

NOTE: Where a KSA exists for multiple certification levels, it is intended that each successive level will possess more in-depth of knowledge of that KSA.

1. Safety

A. Risk Assessment

1. Identify safe work practices with regard to hazards encountered on the job
2. Identify requirements to create a safe work environment on the job site
3. Determine the location and implications of shock and arc-flash protection boundaries

B. Establishing an Electrically Safe Work Condition

1. Apply requirements of NFPA 70E, Article 120: *Establishing an Electrically Safe Work Condition*

C. Lockout/Tagout Procedures

1. Apply OSHA lockout/tagout requirements

D. Personal Protective Equipment (PPE)

1. Recognize and assess Arc-Flash PPE Categories and Shock Protection Boundaries
2. Apply correct use, storage, inspection, and testing of PPE
3. Ensure correct use, storage, inspection, and testing of PPE for the entire crew.

E. Safety Equipment Selection

1. Identify the safety equipment needs of the project (e.g., hotstick, voltage detectors, insulated tools, fire extinguisher, etc.)

F. Confined Space

1. Recognize confined space and its hazards
2. Understand requirements of working within a permit-required confined space (permit, gas monitoring, confined space rescue, etc)
3. Apply OSHA requirements

G. Switching and Temporary Protective Grounding

1. Select and apply correct voltage detection equipment
2. Select and apply correct temporary protective grounding equipment
3. Interpret and apply system switching procedures

H. Incident Energy Analysis

1. Interpret field-marked equipment labels
2. Identify hazards where no field-marked labels exist
3. Demonstrate knowledge of incident energy calculation methods
4. Demonstrate knowledge of the variables involved in incident energy calculation methods
5. Demonstrate understanding of Arc Flash Mitigation techniques and testing requirements in the NEC

I. Codes and Standards

1. Understand scope and application of industry standards and regulatory guidance (e.g., OSHA, ANSI, ASTM, IEEE, NETA, NFPA, NEMA, CSA, etc.)
2. Recognize key provisions of NFPA 70 (e.g., Articles 90, 110.16, 110.21, 225.56, 230.95, 250, and 310) and 70E (i.e. Chapter 1: Safety-Related Work Practices and Chapter 2: Safety-Related Maintenance Requirements, etc.)

Table 3
Level 3 Certified Technician
Knowledge, Skill, and Ability Requirements in Addition to Level 2

3. Apply technical requirements of ANSI/NETA standards
4. Demonstrate an understanding of how manufacturer's instructions and listing and labeling information impacts electrical testing and maintenance

2. Electrical Testing Fundamentals and Theory

A. Fundamentals of Electricity

1. Define and apply terms and concepts from physics, electricity, thermodynamics, and chemistry
2. Recognize and define the standard units used to describe electrical circuits, energy, and power
3. Identify series and parallel circuits
4. Recognize the sources and effects of electromagnetic fields
5. Demonstrate knowledge of dielectric properties of various types of insulations
6. Demonstrate knowledge of phasing and phase rotation

B. Electrical Calculations

1. Perform mathematical calculations utilizing algebra, geometry, and trigonometry
2. Apply fundamental electrical laws to circuits (e.g., Ohm's law, Kirchhoff's law, etc.)
3. Apply conversions between units (e.g., microhms to milliohms, horsepower to kilowatts, etc.)

C. AC and DC Circuits

1. Calculate ac circuits with RLC components
2. Calculate equivalent resistance of and power consumed by resistive circuits
3. Calculate equivalent capacitance and inductance
4. Apply methods for measuring and calculating impedance and power
5. Demonstrate knowledge of complex ac and dc circuits (e.g., voltage drop, current flow, power, reactance, etc.)
6. Demonstrate knowledge of phase angle, power factor, and vectors

D. Insulation Testing

1. Employ methods and procedures for insulation tests and analyze results
2. Demonstrate understanding of Partial Discharge Testing in accordance with ANSI/NETA standards

E. Resistance Testing

1. Employ methods and procedures for resistance tests and analyze results (e.g., winding resistance, contact resistance, bolted connections, etc.)

F. Thermographic Survey

1. Employ methods and procedures for thermographic surveys and analyze results (e.g., emissivity, reflection, delta-T, etc.)

G. Current Testing

1. Employ methods and procedures for current tests and analyze results (e.g., overcurrent pickup, circuit burden, voltage drop, etc.)

H. System Tests, Analysis, and Operation

1. Interpret electrical one-line diagrams
2. Interpret ac and dc schematic diagrams, ac three-line diagrams, and connection and interconnection drawings
3. Identify electrical symbols and ANSI device numbers

Table 3
Level 3 Certified Technician
Knowledge, Skill, and Ability Requirements in Addition to Level 2

4. Interpret time-current curves
5. Recognize essential components of short-circuit and coordination studies (e.g., cable size and length, available protective device settings, raceway type, utility fault contribution, etc.)
6. Apply data extracted from short-circuit and coordination studies
7. Recognize cause and effects of power system harmonics
8. Demonstrate understanding of ANSI/NETA MTS Appendix C Frequency of Power System Studies

3. Component Testing

A. Switchgear and Switchboard Assemblies

1. Identify properties, types, and applications
2. Apply visual and mechanical inspections
3. Employ methods and procedures for electrical tests
4. Evaluate test results

B. Transformers

1. Identify properties, types, and applications
2. Identify properties, types, and applications of auxiliary devices
3. Apply visual and mechanical inspections
4. Employ methods and procedures for electrical tests
5. Evaluate test results

C. Cables

1. Identify properties, types, and applications
2. Identify properties, types, and applications of accessories
3. Apply visual and mechanical inspections
4. Employ methods and procedures for electrical tests
5. Evaluate test results
6. Demonstrate knowledge of cable fault locating

D. Metal-Enclosed Busways

1. Identify properties, types, and applications
2. Apply visual and mechanical inspections
3. Employ methods and procedures for electrical tests
4. Evaluate test results

E. Switches

1. Identify properties, types, and applications
2. Apply visual and mechanical inspections
3. Employ methods and procedures for electrical tests
4. Evaluate test results

F. Circuit Breakers

1. Identify properties, types, and applications
2. Apply visual and mechanical inspections
3. Employ methods and procedures for electrical tests
4. Evaluate test results

G. Circuit Switchers

1. Identify properties, types, and applications
2. Apply visual and mechanical inspections
3. Employ methods and procedures for electrical tests

Table 3
Level 3 Certified Technician
Knowledge, Skill, and Ability Requirements in Addition to Level 2

- 4. Evaluate test results
- H. Network Protectors
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- I. Protective Relays
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- J. Instrument Transformers
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- K. Metering Devices
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- L. Regulating Apparatus
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- M. Grounding Systems
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
 - 5. Employ methods of enhancing effectiveness of grounding systems
- N. Ground-Fault Protections Systems
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- O. Rotating Machinery
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- P. Motor Control Centers and Motor Starters
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results

Table 3
Level 3 Certified Technician
Knowledge, Skill, and Ability Requirements in Addition to Level 2

- Q. Adjustable-Speed Drive Systems
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- R. Direct-Current Systems
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- S. Surge Arresters
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- T. Capacitors and Reactors
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- U. Outdoor Bus Structures
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- V. Emergency Systems
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- W. Automatic Circuit Reclosers and Line Sectionalizers
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- X. Fiber-Optic Cables
 - 1. Recognize and correctly handle fiber-optic cables
 - 2. Apply visual and mechanical inspections
 - 3. Perform fiber tests (loop back test, receive/transmit verification test, loss test)
 - 4. Evaluate test results
- Y. Insulating Liquids and Gases
 - 1. Identify properties, types, and applications
 - 2. Apply sampling procedures
 - 3. Demonstrate understanding of ASTM dielectric breakdown, moisture content, and purity tests
 - 4. Evaluate results of ASTM tests
- Z. Fuses

Table 3
Level 3 Certified Technician
Knowledge, Skill, and Ability Requirements in Addition to Level 2

1. Identify properties, types, and applications
2. Apply visual and mechanical inspections
3. Employ methods and procedures for electrical tests
4. Evaluate test results

AA. Electric Vehicle Charging Systems

1. Identify properties, types, and applications
2. Apply visual and mechanical inspections
3. Employ methods and procedures for electrical tests
4. Evaluate test results

4. Systems and Commissioning

A. Troubleshooting

1. Investigate power or protective system faults or malfunctions to determine cause and corrective action required

B. SCADA

1. Recognize basic requirements, equipment, and configuration of SCADA/DCS systems

C. Monitoring/DCS Systems

1. Apply procedures for microprocessor-based power monitoring and control systems

D. Functional Testing

1. Employ methods and procedures for system-function tests upon completion of the individual component tests defined, as system conditions allow
2. Apply concepts of functionality for electrical systems

Table 4
Level 4 Certified Senior Technician
Knowledge, Skill, and Ability Requirements in Addition to Levels 2 and 3

NOTE: Where a KSA exists for multiple certification levels, it is intended that each successive level will possess more in-depth of knowledge of that KSA.

1. Safety

A. Risk Assessment

1. Identify safe work practices with regard to hazards encountered on the job
2. Identify requirements to create a safe work environment on the job site
3. Determine the location and implications of shock and arc-flash protection boundaries

B. Establishing an Electrically Safe Work Condition

1. Apply requirements of NFPA 70E, Article 120: *Establishing an Electrically Safe Work Condition*

C. Lockout/Tagout Procedures

1. Apply OSHA lockout/tagout requirements for complex, multi-source, and network systems

D. Personal Protective Equipment (PPE)

1. Recognize and assess Arc-Flash PPE Categories and Shock Protection Boundaries
2. Apply correct use, storage, inspection, and testing of PPE personally and for entire crew

E. Safety Equipment Selection

1. Identify the safety equipment needs of the project (e.g., hotstick, voltage detectors, insulated tools, fire extinguisher, etc.)

F. Confined Space

1. Recognize confined space and its hazards
2. Understand requirements of working within a permit-required confined space (permit, gas monitoring, confined space rescue, etc)
3. Apply OSHA requirements

G. Switching and Temporary Protective Grounding

1. Select and apply correct voltage detection equipment
2. Select and apply correct temporary protective grounding equipment
3. Interpret and apply system switching procedures for complex, multi-source, and network systems

H. Incident Energy Analysis

1. Interpret field-marked equipment labels
2. Identify hazards where no field-marked labels exist
3. Demonstrate knowledge of incident energy calculation methods
4. Demonstrate knowledge of the variables involved in incident energy calculation methods
5. Demonstrate advanced understanding of arc energy reduction techniques and testing requirements in the NEC

I. Codes and Standards

1. Understand scope and application of industry standards and regulatory guidance (e.g., OSHA, ANSI, ASTM, IEEE, NETA, NFPA, NEMA, CSA, etc.)

Table 4
Level 4 Certified Senior Technician
Knowledge, Skill, and Ability Requirements in Addition to Levels 2 and 3

2. Recognize key provisions of NFPA 70 (e.g., Articles 90, 110.16, 110.21, 225.56, 230.95, 250, and 310) and 70E (i.e., Chapter 1: Safety-Related Work Practices and Chapter 2: Safety-Related Maintenance Requirements, etc.)
3. Apply technical requirements of ANSI/NETA standards
4. Demonstrate an understanding of how manufacturer's instructions and listing and labeling information impacts electrical testing and maintenance

2. Electrical Testing Fundamentals and Theory

A. Fundamentals of Electricity

1. Define and apply terms and concepts from physics, electricity, thermodynamics, and chemistry
2. Recognize and define the standard units used to describe electrical circuits, energy, and power
3. Identify series and parallel circuits
4. Recognize the sources and effects of electromagnetic fields
5. Demonstrate knowledge of dielectric properties of various types of insulations
6. Demonstrate knowledge of phasing and phase rotation

B. Electrical Calculations

1. Perform mathematical calculations utilizing algebra, geometry, and trigonometry
2. Apply fundamental electrical laws to complex circuits (e.g., Ohm's law, Kirchhoff's law, etc.)
3. Apply conversions between units (e.g., microhms to milliohms, horsepower to kilowatts, etc.)

C. AC and DC Circuits

1. Calculate complex ac circuits with RLC components
2. Calculate equivalent impedance of and power consumed by complex circuits
3. Calculate equivalent capacitance and inductance
4. Apply methods for measuring and calculating impedance and power
5. Demonstrate knowledge of complex ac and dc circuits (e.g., voltage drop, current flow, power, reactance, etc.)
6. Demonstrate knowledge of phase angle, power factor, and vectors

D. Insulation Testing

1. Employ methods and procedures for insulation tests and analyze results
2. Demonstrate advanced understanding of Partial Discharge Testing in accordance with industry standards

E. Resistance Testing

1. Employ methods and procedures for resistance tests and analyze results (e.g., winding resistance, contact resistance, bolted connections, etc.)

F. Thermographic Survey

1. Employ methods and procedures for thermographic surveys and analyze results (e.g., emissivity, reflection, delta-T, etc.)

G. Current Testing

1. Employ methods and procedures for current tests and analyze results (e.g., overcurrent pickup, circuit burden, voltage drop, etc.)

H. System Tests, Analysis, and Operation

1. Interpret electrical one-line diagrams

Table 4
Level 4 Certified Senior Technician
Knowledge, Skill, and Ability Requirements in Addition to Levels 2 and 3

2. Interpret ac and dc schematic diagrams, ac three-line diagrams, and connection and interconnection drawings
3. Identify electrical symbols and ANSI device numbers
4. Interpret time-current curves
5. Recognize essential components of short-circuit and coordination studies (e.g., cable size and length, available protective device settings, raceway type, utility fault contribution, etc.)
6. Apply data extracted from short-circuit and coordination studies
7. Recognize cause and effects of power system harmonics
8. Demonstrate advanced understanding of industry standards as they apply to frequency of power system studies
9. Demonstrate advanced understanding of NERC regulations as they apply to frequency of power system studies

3. Component Testing

A. Switchgear and Switchboard Assemblies

1. Identify properties, types, and applications
2. Apply visual and mechanical inspections
3. Employ methods and procedures for electrical tests
4. Evaluate test results

B. Transformers

1. Identify properties, types, and applications
2. Identify properties, types, and applications of auxiliary devices
3. Apply visual and mechanical inspections
4. Employ methods and procedures for electrical tests
5. Evaluate test results

C. Cables

1. Identify properties, types, and applications
2. Identify properties, types, and applications of accessories
3. Apply visual and mechanical inspections
4. Employ methods and procedures for electrical tests
5. Evaluate test results
6. Demonstrate knowledge of cable fault locating techniques

D. Metal-Enclosed Busways

1. Identify properties, types, and applications
2. Apply visual and mechanical inspections
3. Employ methods and procedures for electrical tests
4. Evaluate test results

E. Switches

1. Identify properties, types, and applications
2. Apply visual and mechanical inspections
3. Employ methods and procedures for electrical tests
4. Evaluate test results

F. Circuit Breakers

1. Identify properties, types, and applications
2. Apply visual and mechanical inspections
3. Employ methods and procedures for electrical tests

Table 4
Level 4 Certified Senior Technician
Knowledge, Skill, and Ability Requirements in Addition to Levels 2 and 3

- 4. Evaluate test results
- G. Circuit Switchers
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- H. Network Protectors
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- I. Protective Relays
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- J. Instrument Transformers
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- K. Metering Devices
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- L. Regulating Apparatus
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- M. Grounding Systems
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
 - 5. Employ methods of enhancing effectiveness of grounding systems
- N. Ground-Fault Protections Systems
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- O. Rotating Machinery
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests

Table 4
Level 4 Certified Senior Technician
Knowledge, Skill, and Ability Requirements in Addition to Levels 2 and 3

- 4. Evaluate test results
- P. Motor Control Centers and Motor Starters
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- Q. Adjustable-Speed Drive Systems
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- R. Direct-Current Systems
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- S. Surge Arresters
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- T. Capacitors and Reactors
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- U. Outdoor Bus Structures
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- V. Emergency Systems
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- W. Automatic Circuit Reclosers and Line Sectionalizers
 - 1. Identify properties, types, and applications
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for electrical tests
 - 4. Evaluate test results
- X. Fiber-Optic Cables
 - 1. Recognize and correctly handle fiber-optic cables
 - 2. Apply visual and mechanical inspections
 - 3. Employ methods and procedures for optical tests
 - 4. Evaluate test results

Table 4
Level 4 Certified Senior Technician
Knowledge, Skill, and Ability Requirements in Addition to Levels 2 and 3

Y. Insulating Liquids and Gases

1. Identify properties, types, and applications
2. Apply sampling procedures
3. Employ methods and procedures for field testing of insulating oils and gases
4. Evaluate test results
5. Recommend corrective actions and remediation techniques

Z. Fuses

1. Identify properties, types, and applications
2. Apply visual and mechanical inspections
3. Employ methods and procedures for electrical tests
4. Evaluate test results

AA. Electric Vehicle Charging Systems

1. Identify properties, types, and applications
2. Apply visual and mechanical inspections
3. Employ methods and procedures for advanced electrical tests
4. Evaluate results of advanced electrical tests

4. Systems and Commissioning

A. Troubleshooting

1. Investigate power or protective system faults or malfunctions to determine cause and corrective action required

B. SCADA

1. Recognize requirements, equipment, and configuration of SCADA/DCS systems

C. Monitoring/DCS Systems

1. Apply procedures for microprocessor-based power monitoring and control systems

D. Functional Testing

1. Employ methods and procedures for complex system-function tests upon completion of the individual component tests defined, as system conditions allow
2. Apply concepts of functionality for complex electrical systems

EXPERIENCE AND TRAINING VERIFICATION BY EMPLOYER

Electrical Test Technician (ETT) – Certified Assistant/Level 2

To be completed by the candidate:

Name _____

Home Address _____

City/State/Zip _____

Phone _____ Fax _____ E-mail _____

Signature _____

To be completed by the employer:

Name _____

Name of Business _____

Address _____

City/State/Zip _____

Phone _____ Fax _____ E-mail _____

I verify that the candidate named above:

- Has a minimum of two years of related experience in the electrical testing industry.
- Has completed a minimum of 40 hours of training* in safety.
- Has completed a minimum of 160 hours of training* related to the certification element categories.**

The certification body retains the right to audit information provided.

Employer - Typed or printed name

Title

Signature

Date

***Training:** *An organized and documented program of activities designed to impart the knowledge and skills required by this standard. On-the-job experience does not qualify as training.*

The training requirement hours for Certified Assistant ETT/Level 2 are in addition to the requirements to attain the Trainee ETT/Level 1 classification.

Acceptable training curricula includes technical schools and factory schools that have provided sufficient documentation that the school is a bona fide training center and that the session was not a sales or product demonstration. This information shall be verified to the certifying body by the employer.

****Safety training is not applicable for this requirement.**

EXPERIENCE AND TRAINING VERIFICATION BY EMPLOYER

Electrical Test Technician (ETT) –Certified Technician Level 3

To be completed by the candidate:

Name _____

Home Address _____

City/State/Zip _____

Phone _____ Fax _____ E-mail _____

ETT Certified Assistant/Level 2 Certification Number _____

Signature _____

To be completed by the employer:

Name _____

Name of Business _____

Address _____

City/State/Zip _____

Phone _____ Fax _____ E-mail _____

I verify that the candidate named above

- Has a minimum of five years of related experience in the electrical testing industry.
- Has completed a minimum of 24 hours of additional training* in safety.
- Has completed a minimum of 240 hours of additional training* related to the certification element categories.**

The certification body retains the right to audit information provided.

Typed or printed name

Title

Signature

Date

***Training:** *An organized and documented program of activities designed to impart the knowledge and skills required by this standard. On-the-job experience does not qualify as training.*

*The training requirement hours for Certified ETT/Level 3 are **in addition to** the requirements to attain the Certified Assistant ETT/Level 2 classification.*

Acceptable training curricula includes technical schools and factory schools that have provided sufficient documentation to the employer that the school is a bona fide training center and that the session was not a sales or product demonstration. This information shall be verified to the certifying body by the employer.

****Safety training is not applicable for this requirement.**

EXPERIENCE AND TRAINING VERIFICATION BY EMPLOYER

Electrical Test Technician (ETT) –Certified Senior Technician Level 4

To be completed by the candidate:

Name _____

Home Address _____

City/State/Zip _____

Phone _____ Fax _____ E-mail _____

ETT Certified Technician Level 3 Certification Number _____

Signature _____

To be completed by the employer:

Name _____

Name of Business _____

Address _____

City/State/Zip _____

Phone _____ Fax _____ E-mail _____

I verify that the candidate named above

- Has a minimum of ten years of related experience in the electrical testing industry.
- Has completed a minimum of 40 hours of additional training* in safety.
- Has completed a minimum of 200 hours of additional training* related to the certification element categories.**

The certification body retains the right to audit information provided.

Typed or printed name

Title

Signature

Date

***Training:** *An organized and documented program of activities designed to impart the knowledge and skills required by this standard. On-the-job experience does not qualify as training. The training requirement hours for Certified Senior ETT/Level 4 are **in addition to** the requirements to attain the Certified Assistant ETT/Level 2 and the Certified ETT/Level 3 classifications.*

Acceptable training curricula includes technical and factory schools that have provided sufficient documentation that the school is a bona fide training center and that the session was not a sales or product demonstration. This information shall be verified to the certifying body by the employer.

****Safety training is not applicable for this requirement.**

REFERENCES

The following references include ANSI/NETA standards, which offer additional reference lists of relevant industry standards.

ANSI/NETA Standard for Acceptance Testing Specifications for Electrical Power

Equipment and Systems

InterNational Electrical Testing Association (NETA)

3050 Old Centre Road

Suite 101

Portage, MI 49024

888.300.6382

neta@netaworld.org

www.netaworld.org

ANSI/NETA Standard for Electrical Commissioning Specifications for Electrical Power

Equipment and Systems

InterNational Electrical Testing Association (NETA)

3050 Old Centre Road

Suite 101

Portage, MI 49024

888.300.6382

neta@netaworld.org

www.netaworld.org

ANSI/NETA Standard for Maintenance Testing Specifications for Electrical Power

Equipment and Systems

InterNational Electrical Testing Association (NETA)

3050 Old Centre Road

Suite 101

Portage, MI 49024

888.300.6382

neta@netaworld.org

www.netaworld.org

Principles of Fairness: An Examination Guide for Credentialing & Registration Boards

The Council on Licensure, Enforcement & Regulation (CLEAR)

108 Wind Haven Drive, Suite A

Nicholasville, KY 40356

859.269.1289

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